

IUST - School of Mathematics and Computer Science

Basic Programming (Python) — Task Team Handbook

Prepared by: Negar Rezaei, Lead TA of Task Team — Fall 2025

Preface

In the dynamic ecosystem of academic learning, organization and clarity are the backbone of quality education. The *Task Team* for the **Basic Programming (Python)** course at IUST was founded with one purpose: to ensure that every weekly task — from design to grading — meets the highest standards of **accuracy, fairness, and educational value**.

Our mission is not merely to create assignments, but to design *learning experiences* that challenge, teach, and inspire students. Each role within this team — Task Designer, QA Reviewer, and Scheduler — plays a vital part in upholding that mission. Together, we form a structured yet collaborative unit, where precision meets creativity and integrity drives every decision.

This handbook serves as a comprehensive guide for all members of the Task Team. It defines responsibilities, workflows, and best practices — so that our work remains transparent, efficient, and consistent across the semester. Every page reflects the collective effort, commitment, and professionalism of our TAs.

— Negar Rezaei
Fall 2025 | IUST

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1. Team Overview & Objectives

هدف تیم | Team Purpose

The Task Team exists to produce high-quality weekly programming tasks that align with the course learning goals, ensure fairness in grading, and maintain the integrity of the evaluation process (including cheating detection).

ترکیب تیم | Composition

- Total members: **8 TAs** (Task Team) — تعداد اعضا: ۸
- Sub-groups: **Task Designers** (design tasks), **QA Reviewer** (quality assurance), **Scheduler** (operations & reporting)

اهداف کلیدی | Key Objectives

- Ensure each task is pedagogically aligned with the week's topic.
- Provide clear, unambiguous problem statements and test coverage.
- Maintain reproducible, well-documented task files in GitHub.
- Enforce deadlines and transparent grading workflows in Quera.
- Detect and address academic integrity issues promptly.

2. Roles & Responsibilities — شرح وظائف

Task Designer — طراح تسک

Primary goal: Produce tasks that are pedagogically sound, clear, and testable.

Main responsibilities | وظائف اصلی

- Receive weekly topic and learning goals from Head TA / Lecturer.
- Design 2-4 **tasks per TA** for the assigned week (as per team rule).
- Fill the Task Template (see Section 4) for each task.
- Provide: problem statement, I/O format, examples, constraints, public tests, hidden tests (at least a draft), rubric, hints (optional), and reference/source.
- Perform local preliminary tests (covering normal & edge cases).
- Submit task PR/branch to GitHub in designated folder and notify QA via Trello.
- Cooperate with QA to implement requested fixes promptly.

Deliverables | خروجی ها

- Task folder with: `task_statement.md` (or `.pdf`), `tests/` (public + hidden), `solution.py` (sample), `rubric.md`, `README.md` (source & notes).

Checklist for Task Designer

- ☐ Learning goal stated
- ☐ Difficulty level assigned (Easy/Medium/Hard)
- ☐ Input/output types and formats specified
- ☐ ≥ 2 worked examples with explanations
- ☐ Public tests exhaustive for basic cases
- ☐ Hidden tests planned for edge + stress cases
- ☐ Sample solution available and passes all tests locally
- ☐ Source/reference cited

QA Reviewer — بازرس کیفیت (QA)

Primary goal: Ensure tasks are structurally sound, technically correct, unambiguous, and aligned to the educational goals.

Main responsibilities | وظایف اصلی

- Review submitted tasks within **48 hours** of TA submission.
- Verify compliance with the Task Template and repository standards.
- Run all provided tests (public + hidden) in pre-release environment (Quera or local autograder).
- Check for English/Persian language clarity and lack of ambiguity in problem statement.
- Ensure edge cases and invalid inputs are handled or explicitly documented as not allowed.
- Tag PRs in Trello as **QA Passed** or **Needs Fix** with specific, actionable feedback.
- Maintain a short weekly summary of recurring issues (e.g., unclear statements, failing hidden tests) and propose corrective training.
- Approve for merge only when task meets standards.

Required skills | مهارت‌های لازم

- Familiarity with `pytest` and Quera autograder basics.
- GitHub + Trello proficiency.
- Strong attention to formatting and ambiguous phrasing.

QA Outputs | خروجی‌های QA

- QA Review Card in Trello with verdict and list of fixes.
- If approved: label `QA Passed` and notify Scheduler/Head TA for scheduling.
- If not approved: `Needs Fix` with exact issues and suggested corrections.

Scheduler — مسئول زمان‌بندی و گزارش

Primary goal: Manage task publication timeline, grade consolidation, protest handling, and weekly reporting.

Main responsibilities | وظایف اصلی

- Coordinate publish times with Head TA and QA.
- Set and announce deadlines in Quera and official channels.
- Enforce late-submission rules and penalties per course policy.
- Close submissions on time per scheduled deadline.

- After task completion, extract Quera grade reports, perform cheating detection checks, consolidate final grades into a central Google Sheet.
- Ensure Quera grades and Google Sheet values match; record `Grade Before` and `Grade After` (post-appeal).
- Triage student appeals and record outcomes. Respond to student within **24 hours** initial reply and finalize within **48 hours** when escalation needed.
- Produce weekly report for Head TA (every Friday night or earlier) including: #submitted, average score, median, number of appeals, cheating incidents, extensions granted.

Scheduler Deliverables | خروجی‌ها

- Google Sheet with final grades and change log.
- Weekly status report (Trello & email to Head TA).
- Records of all deadline extensions, penalties, and grade changes.

3. Workflow — گردش کار رسمی تیم

High-level Process (6 steps)

1. **Task submission by TA** → TA creates PR in `tasks/<week>/` and fills Task Template.
2. **QA review** (technical, structural, content) — complete within 48 hours.
3. **Feedback via Trello** — QA posts `Needs Fix` or `QA Passed`.
4. **Fixes by TA** — TA updates PR; notifies QA.
5. **Final QA approval** → QA merges or gives final sign-off.
6. **Head TA merge & schedule publish** → Scheduler sets publish time in Quera.

Visual/Flow note: For PDF/print, include a simple left-to-right flowchart with icons for Submit → QA → Fix → Approve → Publish → Grade & Report.

Timing rules & SLAs

- QA review must be completed within **48 hours** of submission.
- TAs should respond to `Needs Fix` within **24 hours** (or earlier if deadline approaching).
- Scheduler publishes tasks only after `QA Passed` and Head TA approval (final sign-off).
- Weekly reports by Scheduler due before the next task is published (preferably Friday 23:59).

4. Standards & Templates — قالب‌ها و استانداردها

Task Template (Master)

(Provide this template as `task_template.md` for TAs)

```
# Task Title / عنوان تسک:
[EN] Short descriptive title
[FA] عنوان فارسی (اختیاری)

## Learning Objective / هدف آموزشی:
- [EN] ...
- [FA] ...

## Difficulty Level / سطح سختی:
- Easy / Medium / Hard

## Score / نمره:
- Total: X points
- Breakdown (if applicable): correctness: Y, edge-cases: Z, efficiency: W

## Problem Statement / بیان مسأله:
[EN] Clear English statement...
[FA] متن فارسی (در صورت نیاز)

## Input / Output Format:
- Input: ...
- Output: ...

## Constraints:
- Time limit: X s
- Memory limit: Y MB
-  $n \leq \dots$ 

## Examples (with explanation) / مثال‌ها:
1) Input: ...
   Output: ...
   Explanation: ...

## Tests:
- Public Tests:
  - test1: description
  - test2: ...
- Hidden Tests:
  - hidden1: edge case
  - hidden2: stress case

## Rubric / معیار نمره‌دهی:
- Correctness: ... / X points
- Edge cases coverage: ... / Y points
- Format & constraints: ... / Z points

## Hints (optional) / راهنما:
```

- Hint 1: ...
- Hint 2: ...

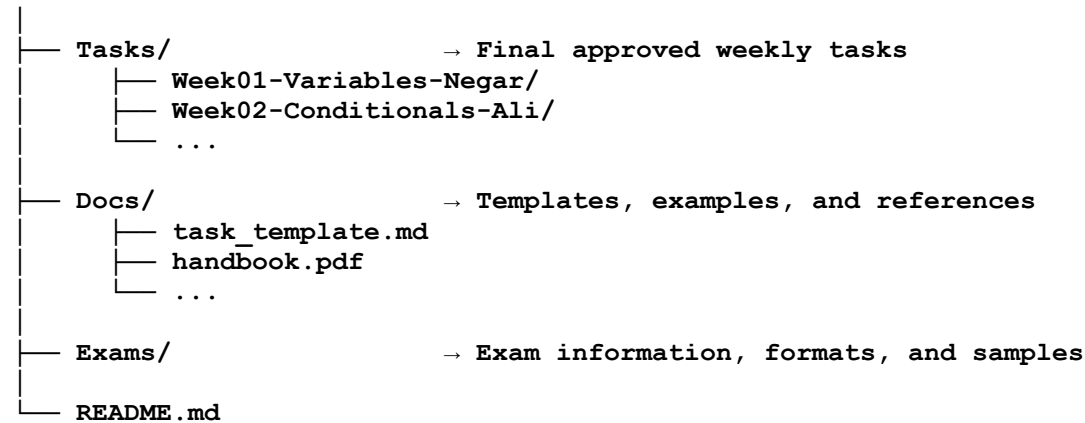
Sample Solution / حل نمونه:
- solution.py (attached)

References / منابع:
- Source: ...
- Notes: ...

File & Repo Naming Convention — قالب نام‌گذاری فایل/شاخه

Consistency is mandatory for automation and CI. Example convention:

Repository root: CS-BP-Tasks-Fall2025
CS-BP-Tasks-Fall2025/



Contribution Guidelines

1 Branches

- Each TA works only in their own branch.
- Recommended naming convention:
- ta/<name>

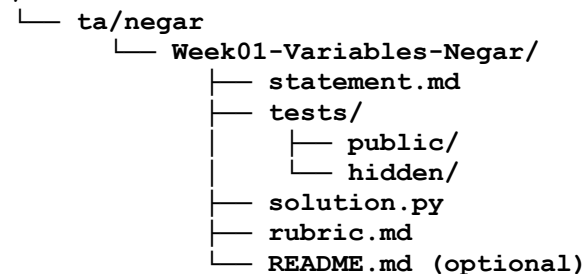
Example: ta/negar

- Never work directly on main.

2 Folder Structure inside Your Branch

When designing your weekly tasks, follow this structure inside your branch:

/CS-BP-Tasks-Fall2025



✱ Folder naming rule:
Week<NN>-<Topic>-<TAName>
Example: Week01-Variables-Negar

3 Committing Files

- Commit early, commit often.
 - Use clear commit messages (e.g., "Added edge test cases for Week01").
 - Keep commits small and focused.
-

4 Pull Requests (PR Workflow)

1. When your weekly tasks are complete and tested locally:
 - Open a Pull Request (PR) from your branch (ta/name) → main.
 2. Assign the PR to the QA Reviewer on GitHub/Trello.
 3. Wait for QA feedback.
 - If "Needs Fix", make the corrections and push updates to your branch.
 4. After "QA Passed", Head TA will approve and merge the PR into main.
 5. Do NOT merge your own PRs.
-

5 After Merge – Folder in Main

When approved and merged, your task will appear in:

/Tasks/Week01-Variables-Negar/

This folder becomes the official copy used for Quera publication.

6 QA Feedback Loop

- QA Reviewer reviews PRs within 48 hours.
 - Feedback will be added both as a Trello update (Needs Fix / QA Passed).
 - You must respond to feedback within 24 hours.
-

7 Head TA Merge & Publication

- After QA approval, Head TA merges into main.
 - Scheduler then coordinates publication in Quera with the same folder and files.
-

8 Guidelines for Edits

- Only modify files in your own branch.
 - Do not alter other TA's folders.
 - Any accidental edits in main will be reverted.
-

9 Automation & CI (Optional Advanced)

For advanced management:

- Enable GitHub Actions CI to auto-test solution.py on each PR.
 - Run pytest tests/ to ensure task integrity before QA review.
 - CI status appears automatically in the PR checks.
-

💬 Contact

For any issue or clarification:

✉ Negar Rezaei – via Telegram @negar636R

📌 Trello Board: CS-BP-Tasks-Fall2025

📍 Repo Maintainers: Task team leader

PR Title format: week-<NN>: <short-title> – TA: <Name>

Rubric Example — نمونه روبریک (نمونه‌ی دقیق)

Total: 10 points

- Correctness (basic cases): **5 points**
 - All public tests pass: 3 points
 - All hidden tests pass: 2 points
- Edge Cases & Input Validation: **3 points**
 - Handles specified invalid inputs or boundary conditions: 3/3
- Performance & Constraints: **1 point**
 - Meets time/memory constraints in hidden stress test: 1/1
- Code Format & Comments: **1 point**
 - Proper formatting, required function signatures, and README: 1/1

Notes: Partial credit must be described explicitly (e.g., if 2 of 3 edge cases covered → 2/3).

Test Design: Public vs Hidden — طراحی تست‌ها

- **Public tests:** cover representative, non-tricky cases for student feedback and practice.
- **Hidden tests:** must include edge cases, invalid input patterns (if allowed), boundary values, and stress tests to ensure performance. Hidden tests must be at least 30–40% of total tests and include at least one test that would fail naive solutions.

Edge case checklist for test designers:

- Min and max values of input sizes
 - Repeated/duplicate inputs
 - Zero / negative values (if applicable)
 - Non-standard spacing or format (if input parsing ambiguous)
 - Large random test for performance (for medium/hard tasks)
-

5. Tools & Communication — ابزارها و کانال‌های ارتباطی

Primary tools | ابزار اصلی

- **Quera:** publish tasks, autograder, collect submissions.
- **GitHub:** repository for task drafts, PRs, CI (optional GitHub Actions).

- **Trello:** task cards, QA statuses (Submitted, Needs Fix, QA Passed, Scheduled, Published).
- **Google Sheets:** consolidated gradebook and appeal logs.
- **Telegram / Official Channel:** announcement and student communication (only official announcements go through Head TA or Scheduler).

Trello Board Structure | ساختار Trello

The **Task Team Trello Board** manages the full lifecycle of each task — from idea to publication.

Each **card** = one task, and each **list** = one workflow stage.

Lists:

-  *Ideas* – new task proposals
-  *To Do (Assigned)* – tasks assigned to TAs
-  *In Progress* – tasks being designed
-  *QA Review* – tasks under QA evaluation
-  *Fixing QA Feedback* – applying revisions
-  *Ready for Merge / Publish* – QA-approved tasks
-  *Published / Done* – final tasks released on Quera

Card Template includes:

Task title, topic, learning goal, assigned TA, difficulty, checklist (design → QA → publish), GitHub link, and attachments (solution, tests, description).

Roles:

- **Head TA:** assigns tasks, monitors progress
- **Task Designers:** design and update tasks
- **QA Reviewer:** reviews and labels QA status
- **Scheduler:** publishes and logs results

Labels: QA Passed, Needs Fix, Delayed, Published.

All updates, feedback, and communication must occur **inside card comments** for transparency and tracking.

Communication rules | قواعد ارتباطی

- All PRs and major decisions must be logged in Trello.
- For any escalation (cheating, big grade discrepancies), notify LEAD TA OF TASK TEAM immediately via private message.
- Scheduler posts weekly status update to Head TA and channel.

6. Workshops & Training Plan — کارگاه‌ها و آموزش‌ها

Mandatory short workshops for TAs:

1. **Designing Programming Assignments** — best practices, difficulty calibration.
2. **Unit Testing & Pytest** — writing public & hidden tests, local testing.
3. **Quera & Autograder** — pre-release tests, publishing flow.
4. **Plagiarism Detection & Interview Protocols** — how to detect, document, and conduct short interviews.
5. **Trello & GitHub Workflow** — PR process, labels, and automation tips.

Schedule recommendation: run 1 workshop in the first 2 weeks, remaining as just-in-time training before major midterms.

7. Appendices — پیوست‌ها

Appendix A — Sample Task (Fully Filled Example)

Title: Compute Average of Positive Integers / محاسبه میانگین اعداد مثبت

Objective: Practice loops, conditional filters, and float formatting.

Difficulty: Easy

Score: 5 points (Correctness 4 / Format 1)

Problem Statement (EN):

Given a list of integers, compute the average of positive integers only. If there are no positive integers, print 0.00.

ورودی (FA):

عدد صحیح از ورودی خوانده می‌شود n (طول لیست) و سپس n اول یک عدد

Input:

n
 $a_1 a_2 \dots a_n$

Output:

Print the average of positive integers with two decimal places.

Examples:

Input:

```
5
-1 2 3 0 -5
```

Output:

```
2.50
```

Explanation: positive numbers are 2 and 3 $\rightarrow$ average $(2+3)/2 = 2.5 \rightarrow$ print 2.50

Public tests: basic small lists, single positive, no positives.

Hidden tests: large n (100k), all positives, mix with zeros, negative-only.

Rubric (5 points): correctness (3), edge-case handling (1), output format (2 decimal places) (1).

Sample solution: `solution.py` (attached), passes all tests.

Appendix B — Cheating Detection Protocol (Short)

1. **Initial detection:** Scheduler or QA flags suspicious similarity or abnormal patterns in Quera submissions.
 2. **Document evidence:** Save code copies, timestamps, and test-case failure patterns.
 3. **Interview step:** TA contacts student with scheduled short interview; record the interview notes.
 4. **Escalation:** If unresolved, report to Head TA with all evidence. Head TA determines penalty per course policy.
 5. **Record:** Maintain `cheating_log` in Google Sheet with final decision.
-

Appendix C — Quick Checklists

Before submitting a task (Task Designer)

- ☐ Template filled fully
- ☐ Local tests pass (public + hidden draft)
- ☐ Solution included and documented
- ☐ File & folder names per convention
- ☐ PR created & Trello card updated

Before approving (QA)

- ☐ Problem statement unambiguous
- ☐ Examples help clarify input/output
- ☐ Public tests cover basic functionality
- ☐ Hidden tests cover edge & stress cases
- ☐ Solution runs and matches documented outputs
- ☐ Rubric is clear and gives partial credit rules

Scheduler weekly duties

- ☐ Publish tasks on schedule
- ☐ Collect Quera reports and check sheet consistency
- ☐ Log grade changes & appeals
- ☐ Report weekly summary to Head TA

Appendix D — Template Rubric Table (for inclusion in PDF)

Criterion	Points	Description
Correctness (public)	3	Passes all public tests
Correctness (hidden)	2	Passes all hidden tests
Edge cases	3	Handles all listed edge cases
Constraints & performance	1	Meets time/memory limits
Formatting & spec	1	Output format exactly as required
Total	10	

Closing Notes — نکات نهایی (Executive summary)

- این دفترچه یک سند عملیاتی و اجرایی است؛ توصیه می‌شود همه اعضای تیم آن را مطالعه کنند و در جلسات هفتگی مورد ارزیابی قرار گیرد.
- قوانین زمان‌بندی و ۴۸ ساعته QA الزامی است؛ عدم رعایت باید با Head TA مطرح شود.
- حفظ مستندسازی (GitHub, Trello, Sheets) برای شفافیت و اتوماسیون در آینده حیاتی است.